

Detector 3 position misses: reference-tracker error, edges, and the passivated area

(June cosmic bench, **M3 v2** basis — `g_det3_wknd`, cross-checked on `sat_det3`)

MX17 cosmic-bench QA

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Supersedes the pre-v2 `main.pdf` in this directory (which used the $\chi^2 < 20$ ray recipe and reported a 13.7% edge-concentrated `reco_far` tail). On the current v2 recipe ($\chi^2 < 5$, `NClus` ≥ 3) that tail is halved, is *not* edge-enhanced, and is roughly half attributable to the reference tracker. This note answers three questions the owner asked: (1) are the > 2 mm misses the *reference tracker's* fault? (2) do they correlate with detector *edges*? (3) is the ~ 2 cm *passivated Y* strip properly handled in the efficiency.

1 Setup

For every clean M3 reference muon crossing the det3 active area we classify the detector's response exactly as `09_efficiency_breakdown.py` does (same alignment, active box, and > 50 -strip spark veto removed *before* the reco/hit split): `reco_near` ($|r| \leq 5$ mm), `reco_far` ($|r| > 5$ mm), `spark`, `hit_no_reco`, `no_hit`. The headline run reproduces the June overview PDF exactly: 88.8% `reco_near`, 6.5% `reco_far`, 4.4% `spark`. Of *reconstructed* hits, 93.2% land within 5 mm, core $\sigma(|r| < 15) = 0.63$ mm, median $|r| = 1.01$ mm. Here we additionally record, per crossing, the **M3 reference-track quality** (`Chi2X`, `Chi2Y`, `NClusX/Y`, incidence angle) and map every ray into the **detector-local strip frame** (0–399 mm) via the inverse alignment transform. We look at the miss at the owner's 2 mm scale as well as the 5 mm cut: $|r| > 2$ mm is 21.5% of reconstructed hits, $|r| > 5$ mm is 6.8%.

2 Q1 — Yes, about half the miss is the reference tracker

The position residual $|r|$ rises monotonically with the M3 track χ^2 (Fig. 1a), *even though the ray recipe already requires* $\chi^2 < 5$. Events in the top decile of M3 χ^2 miss by > 2 mm **50%** of the time, versus 18% for the rest (Fig. 1b). The median M3 $\chi^2_{\max}$ is 0.42 for near (≤ 2 mm) hits but 1.23 for > 2 mm misses — a factor ~ 3 .

Attributing each miss (Fig. 1c) with a bad-reference flag ($\chi^2_{\max} \geq 1$) and a discharge flag (multiplicity ≥ 25 strips): of the > 2 mm misses, **49%** sit on a bad reference track *only*, 12% on a discharge only, 8% on both, and 32% on neither (the ordinary $\sim 3\sigma$ resolution shoulder, expected for a 0.63 mm core at 2 mm). For the genuinely-wrong > 5 mm tail the reference contribution is still the single largest (40% bad-M3-only, 28% discharge-only, 18% both, 14% neither). **So a large part of what the efficiency bookkeeping charges to the chamber as a “position miss” is in fact reference-tracker error** — a slightly-bad telescope track that still passes $\chi^2 < 5$, so the M3 projection is off and the chamber's own point is fine.

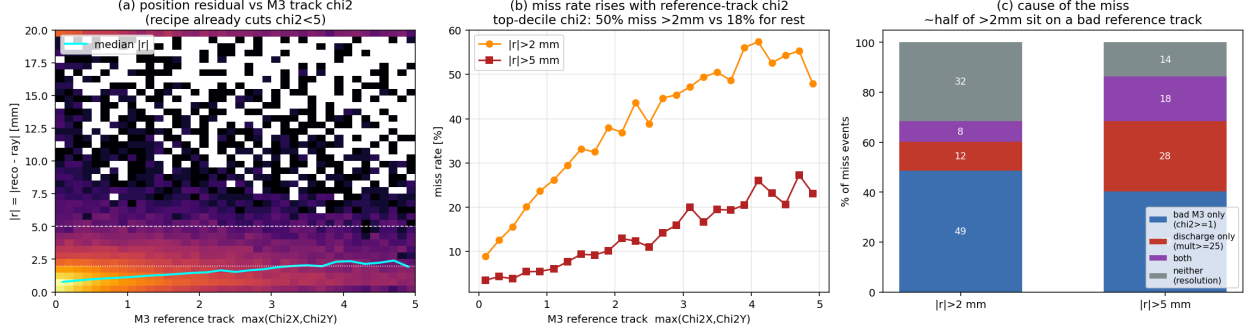


Figure 1: **Is it the reference tracker's fault?** (a) $|r|$ vs M3 track $\max(\chi^2_X, \chi^2_Y)$; the median (cyan) climbs with reference-track χ^2 . (b) miss rate vs M3 χ^2 — top-decile tracks miss $2.7\times$ more often. (c) cause attribution: $\sim$ half of the >2 mm misses occur on an elevated- χ^2 reference track.

3 Intrinsic chamber accuracy

Removing the two pathologies isolates the chamber's true position accuracy (Fig. 2). Of *all* reconstructed hits 93.2% are within 5 mm; requiring a clean reference track ($\chi^2_{\max} < 1$, 68% of events) raises this to **95.8%**, and additionally requiring no sub-spark discharge ($\text{mult} < 25$, together 63% of events) raises it to **98.5%**. The chamber-side residual is the familiar mechanism: the >5 mm miss rate is flat ($\sim 3\%$) up to ~ 25 strips and then climbs steeply toward the 50-strip spark veto (Fig. 2c), i.e. sub-veto discharges throwing a displaced competing cluster. Both effects reproduce on the independent `sat_det3` run (490 V/1000 V): within-5 mm = 93.6/95.9/98.4% for all/clean-M3/clean-both, and top-decile M3- χ^2 misses 48% vs 18%.

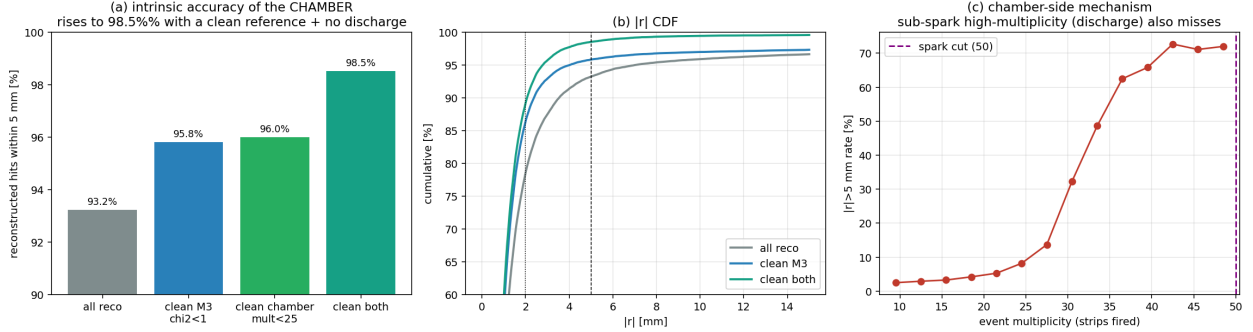


Figure 2: **Intrinsic chamber accuracy.** (a) within-5 mm fraction rises $93.2 \rightarrow 98.5\%$ once reference and discharge pathologies are removed; (b) $|r|$ CDF for the same subsets; (c) chamber-side mechanism — miss rate climbs with multiplicity toward the spark veto.

4 Q2 — No edge enhancement in v2

Mapping the miss rate against distance to the nearest active edge (Fig. 3d), both the >2 mm and >5 mm rates are **flat** across the chamber (6.9% within 10 mm of an edge vs 7.0% in the bulk for >5 mm). The low- X /left-edge concentration reported in the pre-v2 note was a property of the $\chi^2 < 20$ discharge tail; the v2 recipe removes it. The only genuine edge effect is geometric acceptance and the passivated band (Q3), not a mis-reconstruction hot-spot. This transfers to `sat_det3` (edge-distance rates 5.1–6.5%).

5 Q3 — The passivated ~ 2 cm Y strip is real, and already excluded

Projecting every reference ray into the detector-local strip frame and profiling the efficiency (Fig. 3a–c) measures the active area directly:

- **Detector-local Y :** efficiency is dead below $Y \approx 18$ mm and above $Y \approx 380$ mm, flat ($\sim 93\%$ plateau, spark excluded) in between. This is the **passivated band: 18.0 mm at the low edge and 18.7 mm at the high edge** (~ 1.8 cm each, matching the “ ~ 2 cm” recollection). Active Y span = 362 mm of the nominal 399 mm.
- **Detector-local X :** sharp geometric turn-on/off right at 0 and 399 mm, *no* passivation. Active X span = 404 mm (i.e. full width to alignment precision).

The passivation edges come out *identical* on the independently-aligned `sat_det3` run (18.0/18.6 mm, active $Y = [18, 380]$), confirming a real physical property of the chamber rather than an alignment artefact.

Effect on the efficiency. The good news: the current active box in `09_efficiency_breakdown.py` is the 0.5–99.5 percentile of *reconstructed* positions, and since reco only happens in the active Y region, the box self-clips to ~ 362 mm in Y and the passivated dead strips **never enter the denominator**. Recomputing over the explicitly-measured active rectangle ($X \in [-3, 400]$, $Y \in [18, 380]$ mm) gives 88.4% `reco_near` / 6.5% `reco_far` / 4.6% spark — i.e. the headline 88.8% is a genuine *active-area* efficiency, not diluted by passivation. For contrast, if one naïvely used the nominal 40×40 cm the same numerator over the full 399 mm Y would read only $\sim 80\%$; the $\sim 9\%$ passivated Y fraction must not be counted as inefficiency. Recommendation: keep reporting the active-area efficiency, and *document the measured active area* (362×404 mm $\approx 36.2 \times 40.4$ cm, passivated Y band ≈ 18 mm/side) so the denominator is explicit rather than implicit in a percentile.

6 Choosing a stricter reference cut for reprocessing

Because the miss is reference-limited, the natural question is how strict an M3 cut we should adopt. Our sample is already $\chi^2 < 5$, so we scan *stricter* (Fig. 4), applying $\max(\chi_X^2, \chi_Y^2) \leq T$ on both planes and tracking the detector’s core ($|r| < 15$), the `reco_far` rate, and the surviving statistics.

The detector core σ falls monotonically from 0.626 mm ($\chi^2 < 5$) to 0.457 mm ($\chi^2 < 0.3$) with **no plateau** — i.e. reference-tracker error still dominates the measured residual across the whole $\chi^2 < 5$ band, so we “can still see the difference” even at the tightest cut statistics allow. A differential fit ($\sigma^2 = \sigma_{\text{det}}^2 + b\langle\chi^2\rangle$) puts the detector-intrinsic core resolution at $\sigma_{\text{det}} \lesssim 0.3\text{--}0.45$ mm (the cleanest thin bin measures 0.45 mm directly), versus the 0.63 mm we currently quote — the quoted number is inflated by $\gtrsim 40\%$ by reference error. Both runs give identical curves.

Recommendation. Adopt $\boxed{\chi^2 < 1.0 \text{ on both planes} + \text{NClus} = 4}$ as the standard reprocessing cut. It captures the steep part of the improvement (core σ 0.63 $\rightarrow$ 0.47 mm, -25% ; `reco_far` 6.8 $\rightarrow$ 3.3%; within-2 mm 78.5 $\rightarrow$ 89.5%) while retaining $\sim 43\%$ of tracks — ample for the high-statistics June runs (tens of thousands of crossings). If per-bin statistics are tight (fine 2-D maps) fall back to $\chi^2 < 1.0$, $\text{NClus} \geq 3$ (68% retained, σ 0.52 mm). For the dedicated *spatial-resolution* number specifically, push tighter still ($\chi^2 < 0.5$, σ 0.48 mm) or quote the deconvolved intrinsic, since σ has not converged. **Every June analysis (efficiency, spatial resolution, angle) should be re-run on this cut** — see `M3_CUT_AND_ACTIVE_AREA_NOTE.md`.

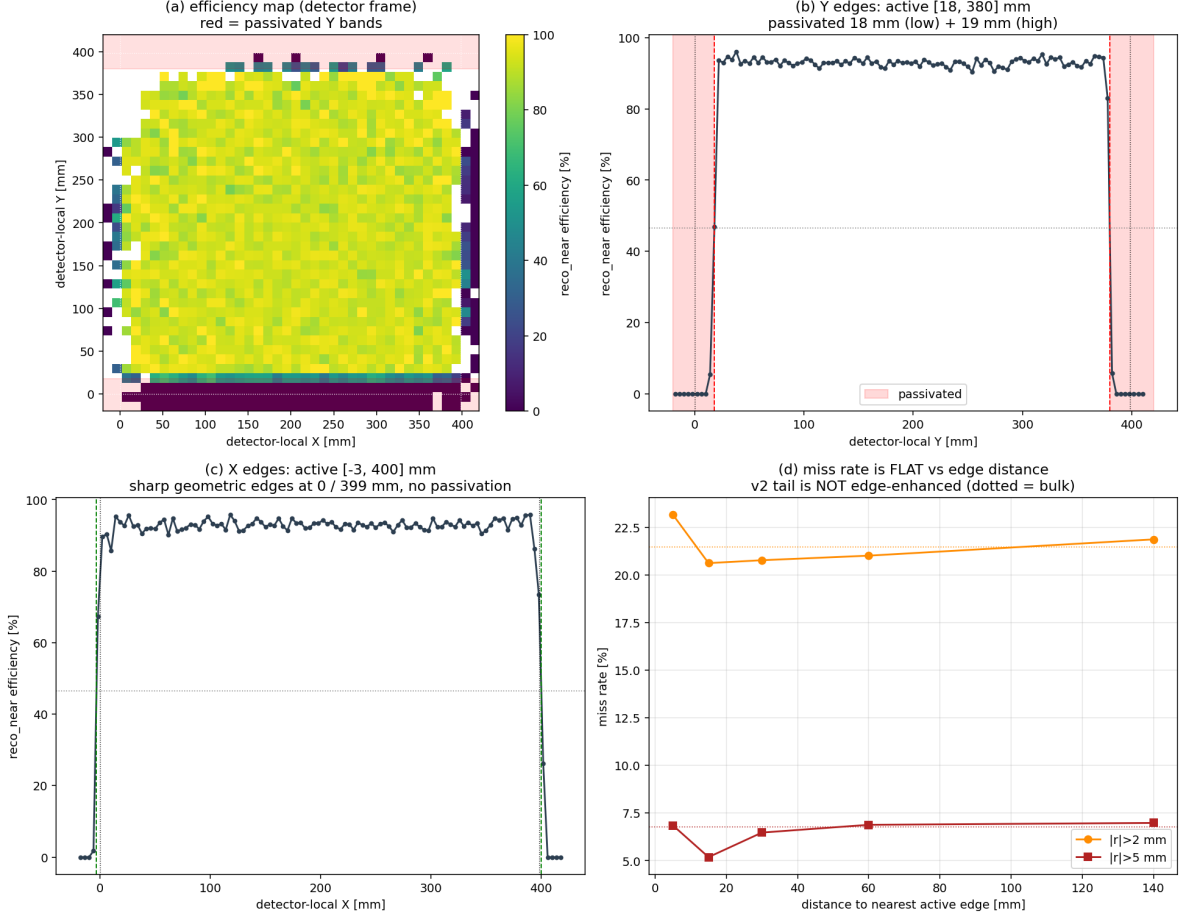


Figure 3: **Edges and passivation (detector-local frame).** (a) efficiency map; the red bands are the passivated Y strips (dead). (b) efficiency vs Y : active [18,380] mm, ~ 18 mm passivated each side. (c) efficiency vs X : sharp geometric edges at 0 / 399 mm, no passivation. (d) miss rate vs distance to nearest active edge — flat, i.e. the $v2$ tail is not edge-driven (dotted = bulk).

7 Takeaways

- The 6.5% `reco_far` (and the broader 21% > 2 mm) is **$\sim$ half reference-tracker error** (elevated- χ^2 M3 tracks that still pass the cut) and **$\sim$ half chamber-side sub-veto discharge**, with a $\sim 30\%$ ordinary-resolution shoulder just past the cut. It is **not** edge-localised and **not** a ray-matching or single-plane fault.
- The chamber's *intrinsic* within-5 mm accuracy is **98.5%** (clean reference + no discharge); the reported 93.2% / core $\sigma = 0.63$ mm are inflated by reference-tracker error ($\gtrsim 40\%$ on σ). The measured residual is reference-limited with no plateau inside $\chi^2 < 5$ — **reprocess everything on $\chi^2 < 1.0$ + NClus = 4** (core $\sigma \rightarrow 0.47$ mm at 43% statistics); go tighter still for the dedicated resolution quote.
- The ~ 2 cm passivated Y band is confirmed and measured (18 mm/side); it is already excluded from the efficiency denominator, so 88.8% is a true active-area number. Document the measured active area ($\approx 36.2 \times 40.4$ cm) explicitly.

Reproduce: `edge_chi2_extract.py` [KEY] (v2 recipe; records M3 track quality + all-crossing categories over a wide

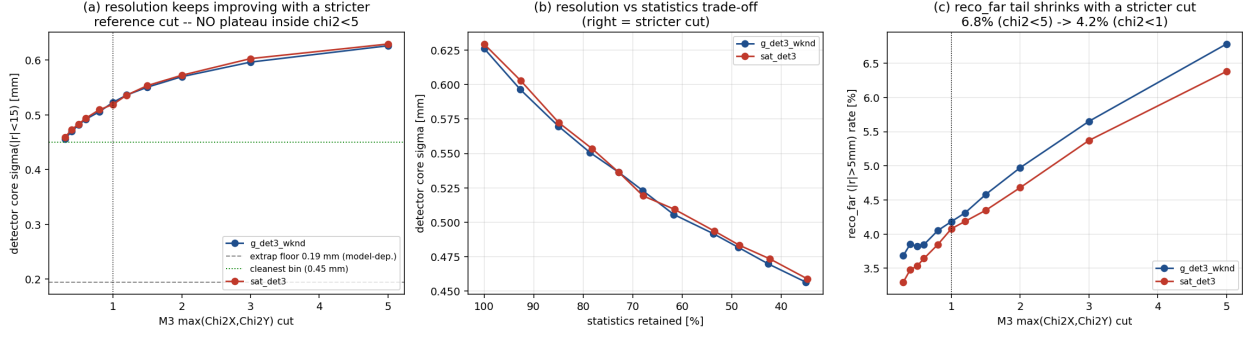


Figure 4: **Stricter reference cut.** (a) detector core σ vs the M3 χ^2 cut — no plateau; the quoted 0.63 mm is reference-limited. (b) σ vs statistics retained (the trade-off; right = stricter). (c) `reco_far` rate shrinks 6.8 $\rightarrow$ 4.2% from $\chi^2 < 5$ to $\chi^2 < 1$.

window) then `edge_chi2_plots.py` [KEY]; numbers in `edge_chi2_meta.<KEY>.json`. Runs: `g_det3_wknd` (headline) and `sat_det3` (490 V/1000 V cross-check). Categorisation matches `09_efficiency_breakdown.py`.